Impact of Climate Change on Vector-Borne Diseases

Hannah Casali
SNCE 110

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Introduction

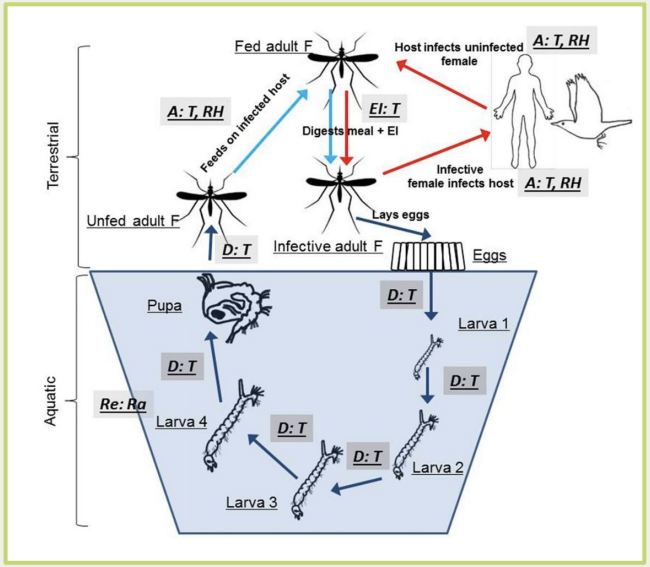
Debate on Climate Change & Disease Risk: Focus on climate-dependent vs. independent factors.

Attribution Challenges: Difficulty isolating climate effects from other influences (e.g., control efforts, socioeconomic factors).

Review Purpose: Analyze predictive methods, evidence, and the role of climate in vector-borne diseases.

Key Findings: Predictions have both over- and underestimated effects, with indirect impacts (e.g., control measures) often ignored.

Emerging Evidence: Increasing but inconclusive evidence linking climate change to some vector-borne diseases.



Why and how do we predict effects of climate change?

Types of Predictive Models:

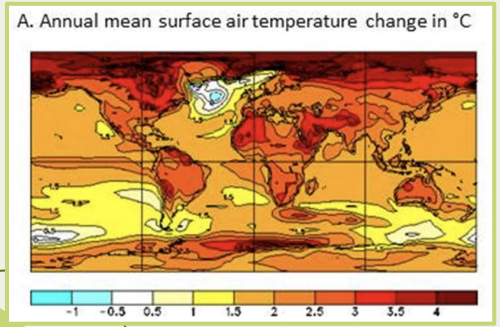
- **Empirical Models:** Use field data to establish climate-disease relationships.
- **Pattern-Matching Models:** Match field data with climate variables to predict disease occurrences.
- **Mathematical Models:** Use mathematical formulations to simulate climate effects on vector populations and disease transmission.

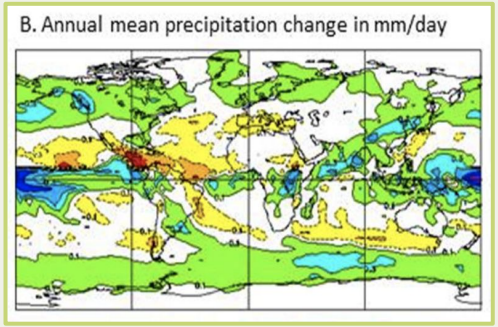
Challenges with Predictive Models:

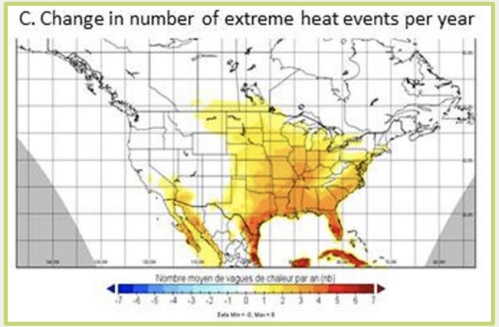
- Predictions can overestimate or underestimate climate impacts.
  - Overestimations: Often ignore climate-independent factors (e.g., control measures).
  - Underestimations: Lack consideration of indirect impacts, like societal adaptations.

Improving Predictions:

- Validation: Using independent data to test model accuracy.
- Uncertainty Analysis: Assessing the uncertainty in model parameters and climate projections.
- Incorporating Non-climate Factors: Including social, economic, and ecological influences in predictions.







Climate Change Effects on Vector Borne Diseases

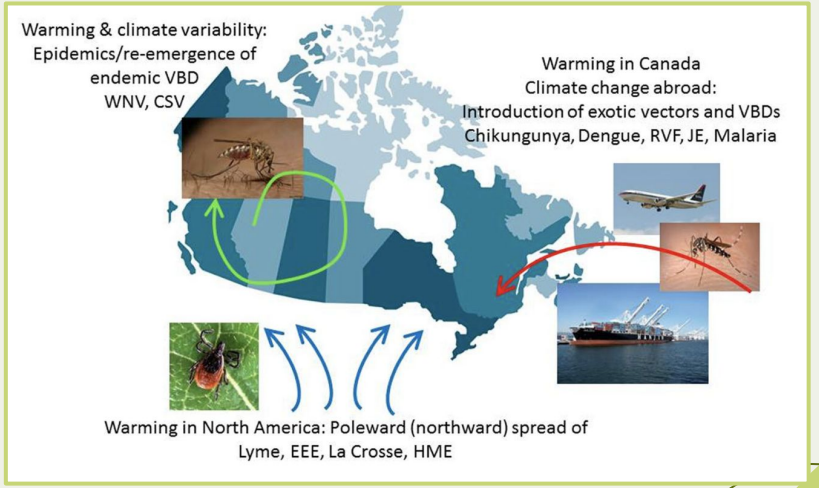
Warming Temperatures : Can alter vector life cycles, increase survival rates, and extend geographical ranges of disease vectors.

Rainfall Changes : Can influence vector breeding habitats, affecting disease transmission cycles.

Increased Frequency of Extreme Weather Events : Affects the spread and intensity of outbreaks, potentially overwhelming public health systems.

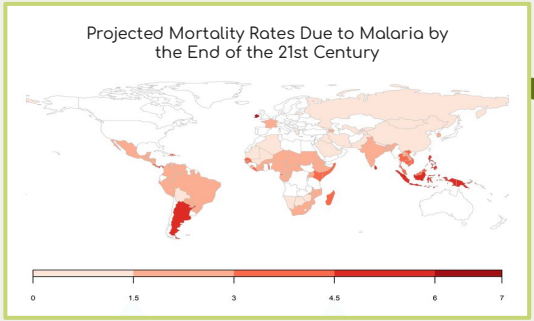
Socioeconomic Impacts : Climate change can lead to economic and social disruptions, weakening control measures and infrastructure for disease prevention.

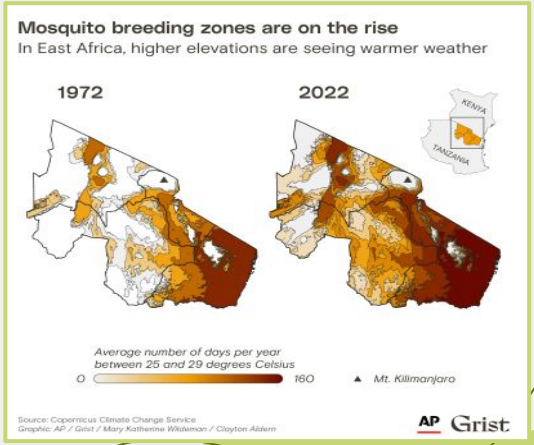
Disrupted Control and Prevention Systems : Climate change may reduce the effectiveness of vector control programs, exacerbating disease risks.



Case Study 1: Malaria

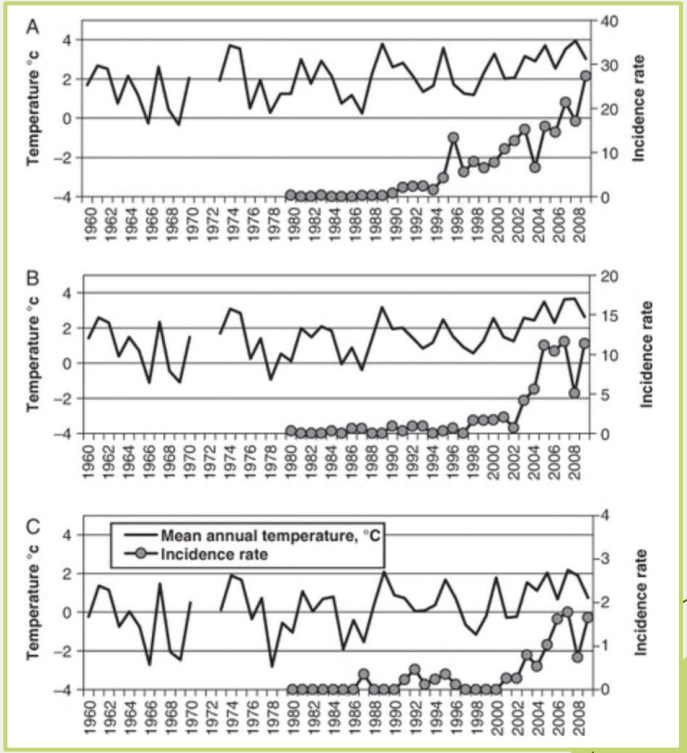
- **Malaria** is highly sensitive to **temperature changes**, affecting mosquito survival and parasite development.
- Temperature influences malaria transmission through **extrinsic incubation periods** and the gonotrophic cycle.
- While some regions show **temperature-related trends** in malaria incidence, confounding factors like **drug resistance** complicate attribution to climate change.





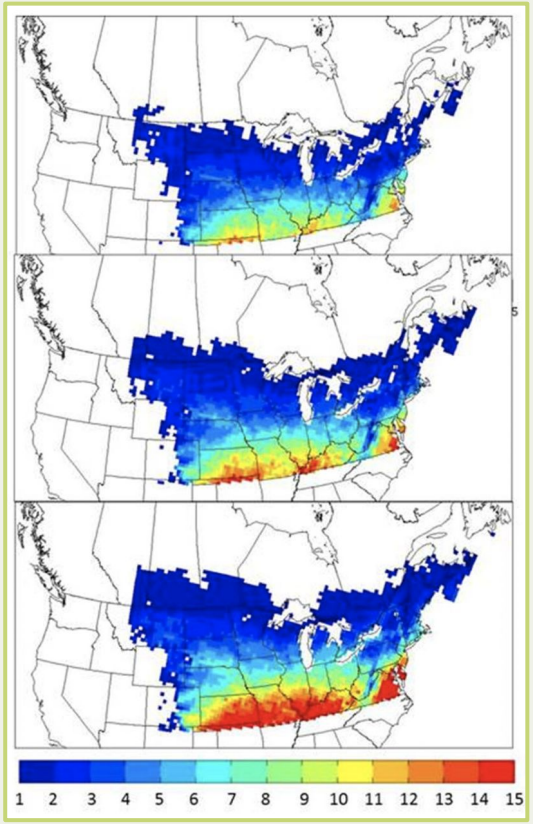
Case Study 2: Tick-Borne Encephalitis (TBE)

- TBE is linked to **tick activity** and **seasonality** , influenced by climate conditions.
- TBE's geographic range may **expand northward** as temperatures **rise**, but other factors such as land use and human behavior are also important.
- Evidence of climate change impact includes **altered tick distribution** and **seasonal patterns.**



Case Study 3: Lyme Disease

- **Lyme disease** incidence is linked to **tick vectors** , particularly *Ixodes scapularis* and *Ixodes ricinus*.
- Rising temperatures **shorten tick life cycles** and allow ticks to **expand their range** .
- **Evidence of Effects:** Lyme disease expansion is clear in regions like Canada and Sweden, where **warming temperatures** have allowed ticks to **survive and thrive in new areas** .



Emerging Threats

Warming climates increase the likelihood of tropical/subtropical diseases like West Nile Virus and chikungunya invading temperate regions.

Examples:

- West Nile Virus: Spread across North America.
- Chikungunya: Outbreak in Italy.
- Dengue and Leishmaniasis: Autochthonous cases reported in Europe.

Climate Change vs. Other Factors:
While climate change increases environmental suitability for disease vectors, globalization and travel have also contributed significantly to these outbreaks.

Challenges in Attribution:
Difficult to determine how much climate change directly influences these invasions, given global travel and trade.

Conclusion

Climate Change's Role: Models must include climate-dependent & independent factors.
Surveillance: Tailored surveillance to risk levels; early warning systems.
Challenges: Data limitations and confounding factors hinder climate attribution.
Public Health Response: Preparedness and control tools are essential.